

Phase 8 — Data Management & Deployment

Goal

Ensure smooth data handling (import/export), prevent duplicates, and safely deploy changes from development to production.

1. Data Import Wizard

Use: For small to medium data volumes (up to 50,000 records) and supported objects like Leads, Contacts, Accounts, and custom objects.

Steps:

1. Setup → Quick Find → **Data Import Wizard** → Launch.
2. Choose Object (e.g., Lead, Contact, or Course__c).
3. Upload CSV file.
4. Map CSV columns → Salesforce fields (e.g., *FirstName*, *LastName*, *Email*, *Preferred_Course__c*).
5. Choose import option (Insert, Update, or Upsert).
6. Start Import.

Result: Records are imported with a detailed **success/failure CSV** report. Errors show issues like missing fields, invalid picklist values, or duplicates.

2. Data Loader

Use: For large datasets (hundreds of thousands/millions) or when populating lookup fields (e.g., Course__c, Batch__c, Enrollment__c).

Steps:

1. Download & install **Data Loader** (from Setup).
2. Login with Salesforce credentials.
3. Choose Insert / Update / Upsert / Delete.
4. Select Object (e.g., Lead, Enrollment__c).
5. Upload CSV → Map CSV columns to **API field names**.
6. Run job.

Result: Large datasets are processed efficiently, supporting lookups via **Record IDs** or **External IDs**.

3. Duplicate Rules

Use: Prevent duplicate records for Leads, Contacts, Tutors, or Students.

Steps:

1. Setup → **Duplicate Management** → Matching Rules → New.
2. Example: Lead Matching Rule = match on **Email**.
3. Create Duplicate Rule → Object = Lead → Action = **Block** or **Allow with Alert**.
4. Repeat for Contacts/Students.

Result: The system blocks or warns users when a duplicate record is created.

4. Data Export & Backup

Use: Create backups before major imports or deployments.

Steps:

1. Setup → **Data Export** → Export Now.
2. Select objects (Lead, Contact, Course__c, Batch__c, Enrollment__c).
3. Salesforce generates a ZIP file with CSVs for download.

Alternative: Data Loader → Export → Save as CSV.

Result: A local backup is available for recovery and auditing.

5. Deployment Options

A. Change Sets (Admin-Friendly)

- Setup → **Outbound Change Sets** → New.
- Add components (custom objects, flows, Apex classes, Lightning Web Components).
- Upload to target org (sandbox → production).
- Monitor in **Deployment Status**.

✓ **Best for admins** with point-and-click deployments.

B. VS Code + Salesforce DX (Developer Approach)

- sf project retrieve → pull metadata.
- sf project deploy start → deploy metadata.
- Works seamlessly with **Git version control**.

✓ **Best for developers** needing modern, scriptable, CI/CD-friendly workflows.

C. ANT Migration Tool (Legacy)

- Java-based command-line tool using build.xml and package.xml.
 - Suitable for scripted CI/CD pipelines.
 - ⚡ Modern projects prefer **SFDX**.
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6. Unmanaged vs Managed Packages

Type	Description	Use Case
Unmanaged	Components editable after install	Internal use, Tutor Match CRM, prototypes
Managed	Locked, versioned, AppExchange-ready	Commercial apps with protected IP

Our Project: Best delivered as an **Unmanaged Package** to allow schools/customers to customize.

7. API Limits & Best Practices

- Use **Data Loader** for >50k records (bulk API).
 - Always **backup data** before imports.
 - Run test imports and deployments in **Sandbox** first.
 - Check **Deployment Status** to verify success.
 - Bulkify imports/updates to avoid hitting governor limits.
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✓ Outcome of Phase 8

- Data (Leads, Courses, Batches, Enrollments) imported and updated.
- Duplicate prevention rules active.

- Backup strategy established.
- Deployment pipeline implemented with **VS Code + SFDX**.
- CRM system ready for safe, production-level use.